

A

a_{11}	a_{12}
a_{21}	a_{22}

 $\times$ B

b_{11}	b_{12}
b_{21}	b_{22}

 $=$ C

c_{11}	c_{12}
c_{21}	c_{22}

$$c_{ij} = \sum_{k=1}^2 a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j}$$